

NCERT Exemplar Solutions For Class 12

Physics

Chapter 2 Electrostatic Potential and Capacitance

Exercise

MULTIPLE CHOICE QUESTIONS- I

2.1 A capacitor of $4\mu F$ is connected as shown in the circuit (Fig. 2.1). The internal resistance of the battery is 0.5Ω . The amount of charge on the capacitor plates will be

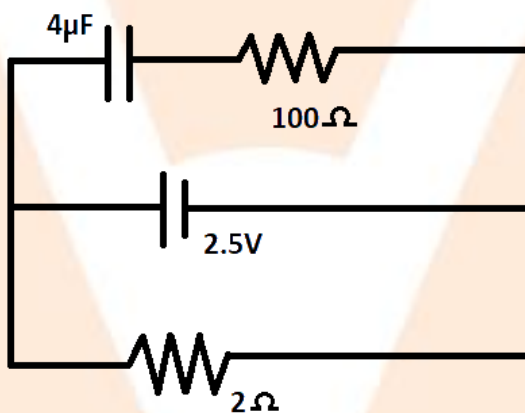


Fig. 2.1

- (a) 0
- (b) $4\mu C$
- (c) $16\mu C$
- (d) $8\mu C$

Ans: (d) $8\mu C$

As the Capacitor offer infinite resistance, the current from cell will not flow across the capacitor branch. So current will flow across 2Ω resistor branch.

$$I = \frac{V}{R+r}$$

$$I = \frac{2.5}{2+0.5}$$

$$I = \frac{2.5}{2.5}$$

$$I = 1\text{Amp}$$

So, the potential difference across 2Ω resistance will be,

$$V = IR$$

$$V = 2 \times 1$$

$$V = 2\text{volt}$$

As the battery, capacitor and 2Ω branches are in parallel. So, the potential difference will remain same across all the three branches.

As the current does not flow through capacitor branch, there will be no potential drop across the 10Ω resistance.

So, the potential difference across $4\mu F$ capacitor is 2volt .

$$q = CV$$

$$q = 4\mu F \times 2$$

$$q = 8\mu C$$

2.2 A positively charged particle is released from rest in a uniform electric field. The electric potential energy of the charge will

(a) Remains a constant because the electric field is uniform.

(b) Increases because the charge moves along the electric field.

(c) Decreases because the charge moves along the electric field.

(d) Decreases because the charge moves opposite to the electric field.

Ans: (c) Decreases because the charge moves along the electric field.

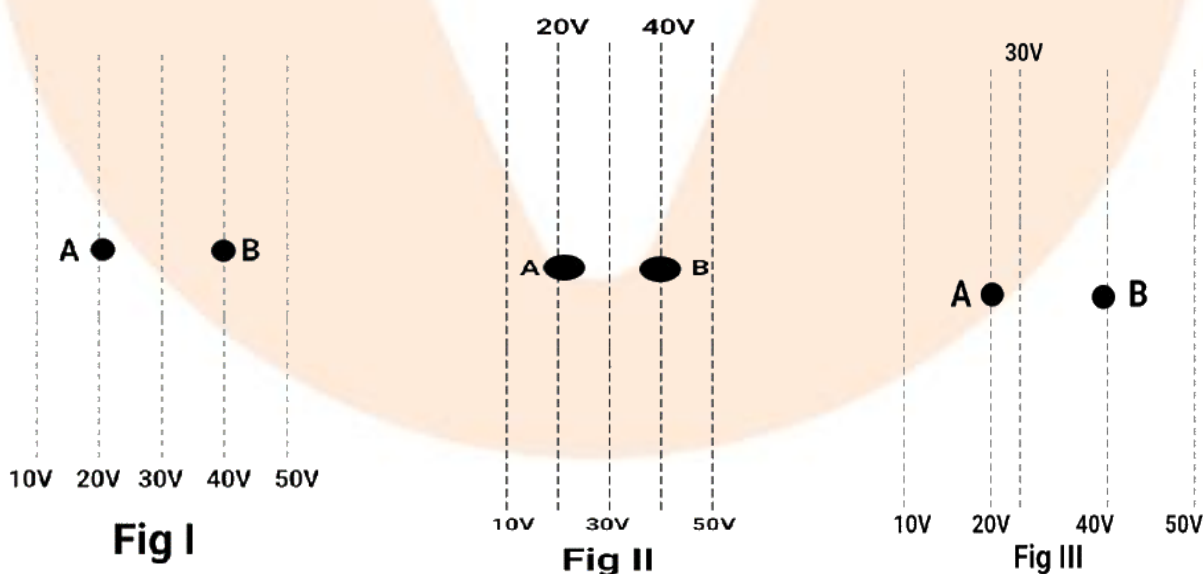
Equipotential surface is always perpendicular to the direction of electric field.

Positive charge experiences the force in the direction of electric field.

When a positive charge is released from rest in uniform electric field, its velocity increases in the direction of electric field. So, the kinetic energy increases, and the potential energy decreases due to law of conservation of energy.

So the potential energy of positively charged particle decreases because the charged particle moves in the direction of field due to the force $q\vec{E}$.

2.3 Figure 2.2 shows some equipotential lines distributed in space. A charged object is moved from point A to point B.



(a) The work done in Fig. (i) is the greatest.

(b) The work done in Fig. (ii) is least.

(c) The work done is same in Fig. (i), Fig. (ii) and Fig. (iii).

(d) The work done in Fig. (iii) is greater than Fig. (ii) but equal to that in Fig. (i)

Ans: (c) The work done is same in Fig. (i), Fig. (ii) and Fig. (iii).

As the potential difference between A and B in all three figures is equal that is 20V, so the work done $\Delta V \cdot q$ by any charge moving from A to B surface will be equal.

2.4 The electrostatic potential on the surface of a charged conducting sphere is 100V . Two statements are made in this regard:

S₁: At any point inside the sphere, electric intensity is zero.

S₂: At any point inside the sphere, the electrostatic potential is 100V .

Which of the following is a correct statement?

(a) S₁ is true but S₂ is false.

(b) Both S₁ & S₂ are false.

(c) S₁ is true, S₂ is also true and S₁ is the cause of S₂.

(d) S₁ is true, S₂ is also true but the statements are independent.

Ans: (c) S₁ is true, S₂ is also true and S₁ is the cause of S₂.

The relation between electric field intensity E and potential V is

$$E = -\frac{dV}{dr}$$

Here, $E = 0$ inside the sphere then $-\frac{dV}{dr} = 0$ i.e. $E = 0$.

$E = 0$ inside charged sphere, the potential is constant or $V = 100$ everywhere inside the sphere and it verifies the shielding effect also.

Hence verifies the option (c).

2.5 Equipotential at a great distance from a collection of charges whose total sum is not zero are approximately

(a) Spheres.

(b) Planes.

(c) Paraboloids

(d) Ellipsoids.

Ans: (a) Spheres

Here we have to find out the shape of equipotential surface. These surfaces are perpendicular to the electric field lines. So there must be an electric field due to the collection of charge.

The algebraic sum of all charges must not be zero. As it is given that the equipotential surface is at a great distance which means that space between the charges is negligible as compared to distance.

So the collection of charges is considered as a point charge.

The electric field lines from a point charge are radial. So the equipotential surface (perpendicular to the field lines) forms a sphere.

It verifies that (a) is the correct Ans:

2.6 A parallel plate capacitor is made of two dielectric blocks in series. One of the blocks has thickness d_1 and dielectric constant k_1 and the other has thickness d_2 and dielectric constant k_2 as shown in Fig. 2.3. This arrangement

can be thought as a dielectric slab of thickness $d = (d_1 + d_2)$ and effective dielectric constant k . The k is

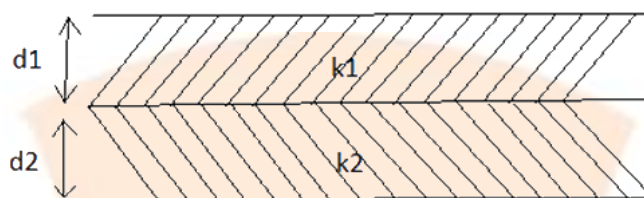


Fig 2.3

(a) $\frac{k_1 d_1 + k_2 d_2}{d_1 + d_2}$

(b) $\frac{k_1 d_1 + k_2 d_2}{k_1 + k_2}$

(c) $\frac{k_1 k_2 (d_1 + d_2)}{(k_1 d_1 + k_2 d_2)}$

(d) $\frac{2k_1 k_2}{k_1 + k_2}$

Ans: (c) $\frac{k_1 k_2 (d_1 + d_2)}{(k_1 d_1 + k_2 d_2)}$

Capacitance of a parallel plate capacitor filled with dielectric of constant k_1 and thickness d_1 is

$$C_1 = \frac{k_1 \epsilon_0 A}{d_1}$$

Similarly for other,

$$C_2 = \frac{k_2 \epsilon_0 A}{d_2}$$

As both the capacitors are in series, the equivalent capacitance C is related as:

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$\frac{1}{C} = \frac{1}{\frac{k_1 \epsilon_0 A}{d_1}} + \frac{1}{\frac{k_2 \epsilon_0 A}{d_2}}$$

$$\frac{1}{C} = \frac{d_1}{k_1 \epsilon_0 A} + \frac{d_2}{k_2 \epsilon_0 A}$$

$$\frac{1}{C} = \frac{1}{\epsilon_0 A} \left(\frac{d_1}{k_1} + \frac{d_2}{k_2} \right)$$

$$\frac{1}{C} = \frac{1}{\epsilon_0 A} \left(\frac{k_2 d_1 + k_1 d_2}{k_1 k_2} \right)$$

$$\text{So } C = \frac{k_1 k_2 \epsilon_0 A}{(k_2 d_1 + k_1 d_2)} \dots\dots\dots \text{eq I}$$

$$C' = \frac{k \epsilon_0 A}{d} \dots\dots\dots \text{eq II}$$

Here $d = (d_1 + d_2)$

So, multiply the numerator and denominator of eq I with $(d_1 + d_2)$,

$$C = \frac{k_1 k_2 \epsilon_0 A}{(k_2 d_1 + k_1 d_2)} \cdot \frac{(d_1 + d_2)}{(d_1 + d_2)}$$

$$C = \frac{k_1 k_2 (d_1 + d_2)}{(k_2 d_1 + k_1 d_2)} \cdot \frac{\epsilon_0 A}{(d_1 + d_2)} \dots\dots\dots \text{eq III}$$

On comparing eq II and eq III, the dielectric constant of new capacitor is:

$$\frac{k_1 k_2 (d_1 + d_2)}{(k_1 d_1 + k_2 d_2)}$$

It verifies that the correct Ans is (c).

MULTIPLE CHOICE QUESTION-II MORE THAN ONE OPTION

2.7 Consider a uniform electric field in the $\hat{z}$ direction. The potential is a constant

- (a) in all space.**
- (b) for any x for a given z .**
- (c) for any y for a given z .**
- (d) on the $x-y$ plane for a given z .**

Ans: (b) for any x for a given z , (c) for any y for a given z and (d) on the $x-y$ plane for a given z .

As we know that equipotential surfaces are perpendicular to the direction of electric field lines.

Here electric field is in $+\hat{z}$ direction.

So, equipotential surfaces will be in the plane perpendicular to z -axis i.e. along $x-y$ plane which includes any x or y axes.

So the Ans (b), (c) and (d) are verified respectively.

2.8 Equipotential surfaces

- (a) are closer in the regions of large electric fields as compared to the regions of lower electric fields.**
- (b) will be more crowded near sharp edges of a conductor.**
- (c) will be more crowded near regions of large charge densities.**
- (d) will always be equally spaced.**

Ans: (a) are closer in the regions of large electric fields as compared to the regions of lower electric fields, (b) will be more crowded near sharp edges of a conductor and (c) will be more crowded near regions of large charge densities.

We know that on any two points of equipotential surface, potential difference is zero or of equal i.e.

$$E = -\frac{dV}{dr}$$

The electric field intensity is inversely proportional to the separation between equipotential surfaces. So the equipotential surfaces are closer in the regions of large electric fields.

Thus, it verifies Ans (a).

The electric field is larger near the sharp edge, due to larger charge density as A is very small i.e.

$$\sigma = \frac{q}{A}$$

So the equipotential surfaces are closer or crowded.

Thus, it verifies Ans (b).

As the electric field $E = \frac{kq}{r^2}$ and potential or field decreases as size of body increases or vice-versa (case of earth), the equipotential surfaces will be more crowded if the charge density $\sigma = \frac{q}{A}$ increases.

Thus, it verifies the Ans (c).

As the equipotential surface depends on distance r by $E = -\frac{dV}{dr}$ and $V = \frac{kq}{r}$.

Equipotential surface depends on charge density at that place which is different at different place, so equipotential surface are not equally spaced all over.

2.9 The work done to move a charge along an equipotential from A to B

(a) cannot be defined as $-\int_A^B E \cdot dl$

(b) must be defined as $-\int_A^B E \cdot dl$

(c) is zero.

(d) can have a non-zero value.

Ans: (c) is zero.

As the potential on equipotential surface does not change

$$\text{So } V_2 - V_1 = 0$$

$$\text{And } W = (V_2 - V_1)q$$

So, work done on moving a charge is zero, verifies Ans (c).

We know the work done by a charge q moving in electric field is

$$dW = F \cdot dl$$

$$\int = \int qE \cdot dl$$

$$W = q \int E \cdot dl$$

So, $W \neq \int E \cdot dl$ or Ans (b) is wrong.

Ans (a) and (b) can be true only when $q = +1C$ which is not given in question.

2.10 In a region of constant potential

(a) the electric field is uniform

(b) the electric field is zero

(c) there can be no charge inside the region.

(d) the electric field shall necessarily change if a charge is placed outside the region.

Ans: (b) the electric field is zero and (c) there can be no charge inside the region.

Constant potential $\Rightarrow dV = 0$ so by relation $E = -\frac{dV}{dr}$, $E = 0$ i.e. the electric field is not uniform which discards the Ans (a) and agree with the Ans (b).

As the potential may also be present outside the charge, there can be no charge inside the region of constant potential.

Thus, this verifies Ans (c).

If a charge is placed outside the region, the potential difference or electric field in the region will change.

Thus, this makes Ans (d) incorrect.

2.11 In the circuit shown in Fig. 2.4 initially key K_1 is closed and key K_2 is open. Then K_1 is opened and K_2 is closed (order is important). [Take Q_1 and Q_2 as charges on

C_1 and C_2 and V_1 and V_2 as voltage respectively].

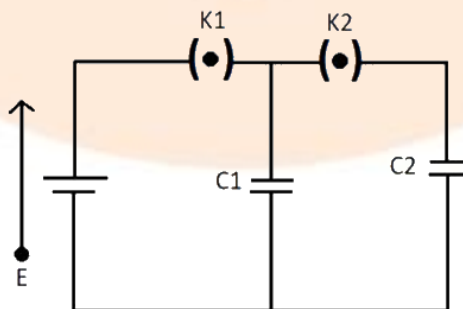


Fig. 2.4

Then

(a) charge on C_1 gets redistributed such that $V_1 = V_2$

(b) charge on C_1 gets redistributed such that $Q_1' = Q_2'$

(c) charge on C_1 gets redistributed such that $C_1V_1 + C_2V_2 = C_1E$

(d) charge on C_1 gets redistributed such that $Q_1' + Q_2' = Q$

Ans: (a) charge on C_1 gets redistributed such that $V_1 = V_2$ and (d) charge on C_1 gets redistributed such that $Q_1' + Q_2' = Q$

When K_1 is closed keeping K_1 open, the capacitor C_1 gets charged by battery of emf E . Now, when K_1 opens C_1 remains charged. When K_2 closes keeping K_1 open, C_2 gets charged by redistribution of charges of C_1 between C_1 and C_2 .

Let the charge on C_1 which is charged by battery be Q , then after redistribution of charge $Q = Q_1' + Q_2'$ by law of conservation of charge.

So the Ans (d) is verified.

As C_1 and C_2 both are in parallel combination, so their potential will be equal i.e. $V_1 = V_2$.

Thus, it verifies the Ans (a).

2.12 If a conductor has a potential $V \neq 0$ and there are no charges anywhere else outside, then

(a) there must be charges on the surface or inside itself.

(b) there cannot be any charge in the body of the conductor.

(c) there must be charges only on the surface.

(d) there must be charges inside the surface.

Ans: (a) there must be charges on the surface or inside itself. (b) there cannot be any charge in the body of the conductor.

As the excess charge can reside only on the surface of a conductor and inside net positive and negative charge is zero.

Any charge can reside inside the hollow shell or body. So this verifies Ans (a) and discards Ans (c).

Inside the solid material of conducting body there is no charge, it comes to outer surface. So this verifies Ans (b) and discards Ans (d).

2.13 A parallel plate capacitor is connected to a battery as shown in Fig. 2.5. Consider two situations:

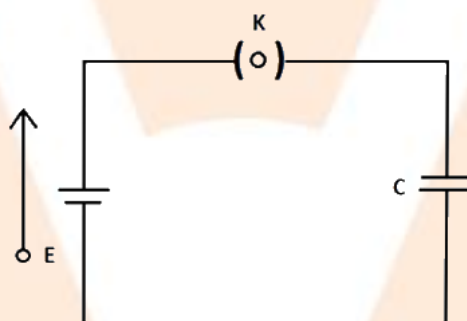


Fig. 2.5

A: Key K is kept closed and plates of capacitors are moved apart using insulating handle.

B: Key K is opened and plates of capacitors are moved apart using insulating handle.

Choose the correct option(s).

(a) In A: Q remains same but C changes.

(b) In B: V remains same but C changes.

(c) In A: V remains same and hence Q changes.

(d) In B: Q remains same and hence V changes.

Ans: (c) In A: V remains same and hence Q changes and (d) In B: Q remains same and hence V changes.

(A) In situation A: When the space between the plates of capacitor increases, the capacitance decreases by relation $C = \frac{K\epsilon_0 A}{d}$, but battery remains same i.e. potential difference across plate remains ' V ' same. So by $Q = CV$ relation, Q also decreases which verify the Ans (c) and discards the Ans (a).

(B) Now for situation B: K is open and capacitance decreases by moving apart the plates of capacitor, so by relation $Q = CV$, here K is open so charge Q remains same in turn V will increase on decreasing C hence Ans (d) is verified.

VERY SHORT ANSWER TYPE QUESTIONS

2.14 Consider two conducting spheres of radii R_1 and R_2 with $R_1 > R_2$. If the two are at the same potential, the larger sphere has more charge than the smaller sphere. State whether the charge density of the smaller sphere is more or less than that of the larger one.

Ans: Let Q_1 and Q_2 be the charges on radii R_1 and R_2 we know that

$$V = \frac{1}{4\pi\epsilon} \frac{Q_1}{R_1} \text{ and } V = \frac{1}{4\pi\epsilon} \frac{Q_2}{R_2}$$

As $V_1 = V_2$, so

$$\Rightarrow \frac{1}{4\pi\epsilon} \frac{Q_1}{R_1} = \frac{1}{4\pi\epsilon} \frac{Q_2}{R_2}$$

$$\Rightarrow \frac{Q_1}{R_1} = \frac{Q_2}{R_2}$$

$$\Rightarrow \frac{Q_1}{Q_2} = \frac{R_1}{R_2}$$

$$\Rightarrow \frac{\sigma_1 \times 4\pi(R_1)^2}{\sigma_2 \times 4\pi(R_2)^2} = \frac{R_1}{R_2}$$

$$\Rightarrow \frac{\sigma_1}{\sigma_2} = \frac{R_2}{R_1}$$

Since $R_1 > R_2$ it follows that

$$\sigma_1 < \sigma_2$$

i.e. the charge density of smaller sphere is less than that of larger sphere.

2.15 Do free electrons travel to region of higher potential or lower potential?

Ans: As free electrons have negative charge so the direction of their flow will be opposite to positive charge i.e. free electrons will move from lower potential to higher potential.

2.16 Can there be a potential difference between two adjacent conductors carrying the same charge?

Ans: If the current flowing through the two conductors is same then they both may be considered in series.

So according to Ohm's law becomes

$$V \propto R$$

i.e. if the resistances (which depends on ρ , l and A) are different then the potential difference will also be different.

So there can be a potential difference between two adjacent conductors carrying the same charge or current if either their length or area of cross-section (A) and resistivity are different.

2.17 Can the potential function have a maximum or minimum in free space?

Ans: In the absence of free space or atmosphere, the phenomenon of electric field or potential leakage is cannot be prevented. Hence, the potential function do not have maximum or minimum in free space.

2.18 A test charge q is made to move in the electric field of a point charge Q along two different closed paths (Fig. 2.6). First path has sections along and perpendicular to lines of electric field. Second path is a rectangular loop of the same area as the first loop. How the work done does is compare in the two cases?

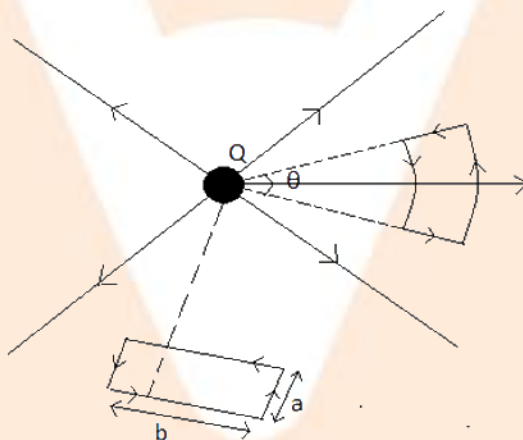


Fig 2.6

Ans: We know that electrostatic work done is conservative. So work done in closed loop is always zero, it does not depend on the nature of closed path.

SHORT ANSWER TYPE QUESTIONS

2.19 Prove that a closed equipotential surface with no charge within itself must enclose an equipotential volume.

Ans: Let us consider that inside the enclosed equipotential surface, potential is not same. Let the potential inside the equipotential surface is different to that on the equipotential surface, causing in a potential gradient $\frac{dV}{dr} = E$. So the electric field which exists inside surface is equal to $E = \frac{dV}{dr}$.

The electric field lines pointing inward or outward from the surface are perpendicular to the equipotential surfaces or the field lines cannot be on the equipotential surface.

The field lines can be on the equipotential surface if field lines can originate from the charge inside, which contradicts the original assumption. Hence, the entire volume inside equipotential surface has no charge.

2.20 A capacitor has some dielectric between its plates, and the capacitor is connected to a DC source. The battery is now disconnected and then the dielectric is removed. State whether the capacitance, the energy stored in it, electric field, charge stored and the voltage will increase, decrease, or remain constant.

Ans: $C = \frac{K\epsilon_0 A}{d}$

As k is positive and more than one, so by removing dielectric slab, and keeping A and d constant, capacitance of capacitor will decrease. When the battery and dielectric slab from capacitor is removed, the charge remains same as it was when battery connected earlier.

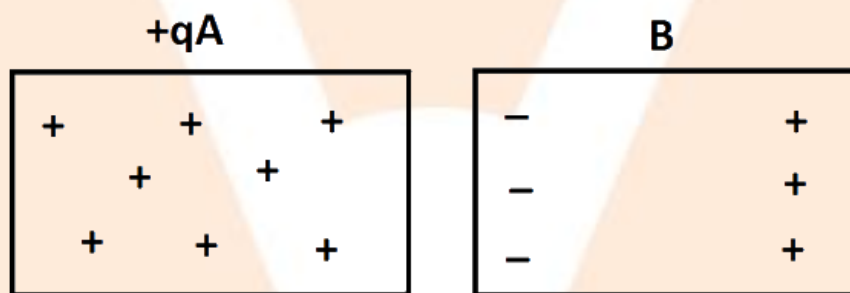
As the energy stored in capacitor is $\frac{q^2}{2C}$. When capacitance C is decreased by removing dielectric slab but q remains same, so the energy stored in capacitor will increase.

We know that $V = \frac{q}{C}$, where q is same and C is decreased so potential will increase.

As $E = \frac{V}{d}$, distance between plates of capacitor is same and potential is increased as discussed above, so electric field between the plates of capacitor will increase.

2.21 Prove that, if an insulated, uncharged conductor is placed near a charged conductor and no other conductors are present, the uncharged body must be intermediate in potential between that of the charged body and that of infinity.

Ans: Consider a charged body (A) (say with positive charge) and an insulated uncharged conductor (B) is placed near the charged conductor (A) as shown in the figure:



As $V = \frac{kq}{r}$ where k and q are constants, so

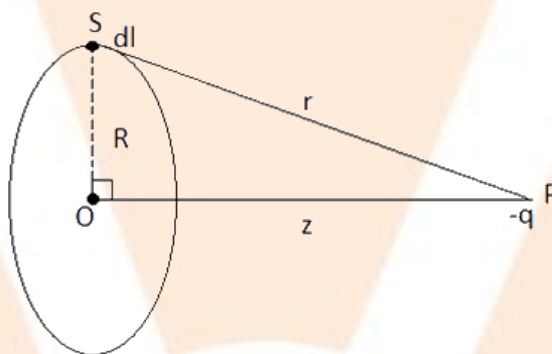
$V \propto \frac{1}{r}$ or infinity, $V \rightarrow 0$

Uncharged conductor is between charged conductor and infinity, so potential decreases from body A to infinity.

So the potential of uncharged body varies between potential of A and infinity.

2.22 Calculate potential energy of a point charge $-q$ placed along the axis due to a charge $+Q$ uniformly distributed along a ring of radius R . Sketch potential energy as a function of axial distance z from the center of the ring. Looking at graph, can you see what would happen if $-q$ is displaced slightly from the center of the ring (along the axis)?

Ans: Let us consider a ring of radius R having charge $+Q$ distributed uniformly over the ring. Also there is a point P at distance a z on the axis passing through center O and perpendicular to plane of the ring.



Again consider an element of ring at S of length dl having charge dq and SP is equal to r . Then the potential energy due to the element dl at P if dq is the charge on element dl of ring is $dV = \frac{-kdq}{r}$ where $k = \frac{1}{4\pi\epsilon_0}$ and as Q is positive charge, so potential due to the charge dq will be negative.

Charge on $2\pi R$ length of the ring $= Q$

Charge on dl length of the ring is $dq = \frac{Q}{2\pi R} dl$

So potential due to element dl at P

$$dV = \frac{-kQdl}{2\pi Rr}$$

Since $dW = dV \cdot q$ and $r = \sqrt{R^2 + z^2}$

$$\text{So, } dW = \frac{-kQqdl}{2\pi R\sqrt{R^2 + z^2}}$$

Integrating both sides, over a ring, we have

$$\int_0^\pi dW = - \int_0^{2\pi R} \frac{kqQdl}{2\pi R\sqrt{R^2 + z^2}}$$

$$W = - \frac{kqQ2\pi R}{2\pi R\sqrt{1 + \frac{z^2}{R^2}}}$$

This work done gets converted into P.E. at P, so P.E. is

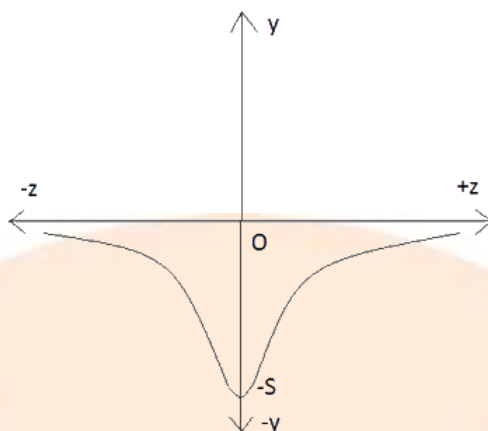
$$V = \frac{-Qq}{4\pi\epsilon_0 R\sqrt{1 + \frac{z^2}{R^2}}}$$

Let $\frac{Qq}{4\pi\epsilon_0 R} = S$ (a new constant)

$$V = \frac{-S}{\left[1 + \frac{z^2}{R^2}\right]^{\frac{1}{2}}}$$

At $z = -\infty$

$$V_2 = \frac{-S}{\left[1 + \frac{z^2}{R^2}\right]^{\frac{1}{2}}}$$



Hence, $z \gg R$

Also, $z^2 \gg R^2$

$$\left(1 + \frac{z^2}{R^2}\right)^{\frac{1}{2}} = \infty$$

$$\text{So, } V_{-z} = \frac{-S}{\infty} \rightarrow 0$$

$$V_{-z} \rightarrow 0^{\infty}$$

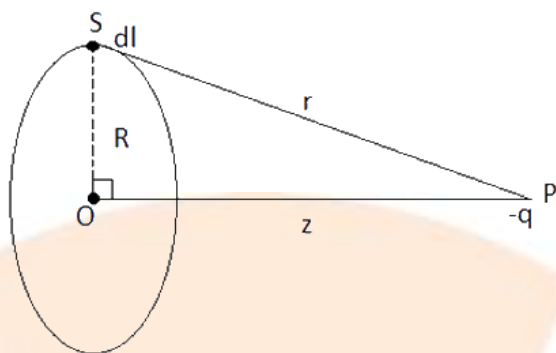
$$V_{+z} \rightarrow 0$$

At $z=0$

$$V = -S$$

2.23 Calculate potential on the axis of a ring due to charge Q uniformly distributed along the ring of radius R.

Ans: Let us consider a ring of radius R having charge $+Q$ distributed uniformly. Also there is a point P at distance z on its axis passing through center O and perpendicular to plane of ring.



Again consider an element on ring at S of length dl having a charge dq and SP is equal to r . Then potential energy due to element dl at P is,

$$dV = \frac{-kdq}{r} \text{ where } k = \frac{1}{4\pi\epsilon_0}$$

Charge on $2\pi R$ length of ring $= Q$

$$\text{Charge on } dl \text{ length of ring} = \frac{Q}{2\pi R} dl$$

So potential due to element dl at P ,

$$dV = \frac{-k \cdot Q \cdot dl}{2\pi R r}$$

Integrating over a ring the potential at P , V_p

$$\int_0^V dV_p = \int_0^{2\pi R} \frac{kQdl}{2\pi R r} \text{ where } r = \sqrt{R^2 + z^2}$$

$$V_p = \frac{kQ2\pi R}{2\pi R \sqrt{R^2 + z^2}}$$

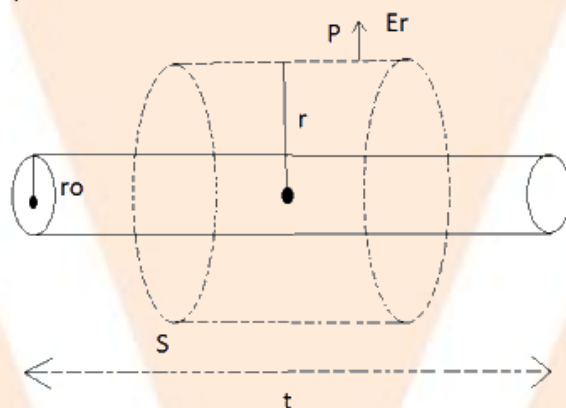
$$V_p = \frac{Q}{4\pi\epsilon_0 \sqrt{R^2 + z^2}}$$

LONG ANSWER TYPE QUESTIONS

2.24 Find the equations of the equipotential surfaces for an infinite cylinder of radius r_0 and carrying a charge of linear density λ .

Ans: Consider a Gaussian cylindrical dotted surface S at a distance r from the center of the cylinder of radius r_0 of the infinite length.

The electric field lines are radial and perpendicular to the surface.



Let electric field intensity on Gaussian surface at P is E , and total charge q on cylinder will be $q = \lambda l$.

So, by Gauss's law,

$$\oint_S E_r ds = \frac{\lambda l}{\epsilon} \Rightarrow [E_r \cos \theta]_0^{2\pi l} = \frac{\lambda l}{\epsilon_0}$$

$$E, 2\pi r l \cos 90^\circ = \frac{\lambda l}{\epsilon} \quad [\angle \theta \text{ is between } E_r \text{ and curved surface of dotted cylinder is } 90^\circ]$$

$$E_r = \frac{\lambda}{2\pi r \epsilon_0}$$

We know that electric field E_r at distance r from center of cylinder $E_r = \frac{-dV}{dr}$.

So the potential difference dV at distance r_0 and r from the center of cylinder is,

$$dV = E_r \cdot dr \left[\because E = \frac{-dV}{dr} \right]$$

$$V(r) - V(r_0) = -\int E_r \cdot dr$$

$$V(r) - V(r_0) = -\int_{r_0}^r \frac{\lambda}{2\pi\epsilon_0 r} \cdot dr = \frac{-\lambda}{2\pi\epsilon_0} \int_{r_0}^r \frac{dr}{r} = \frac{-\lambda}{2\pi\epsilon_0} [\log_e r]_{r_0}^r$$

$$V(r) - V(r_0) = \frac{-\lambda}{2\pi\epsilon_0} [\log_e r - \log_e r_0] = \frac{-\lambda}{2\pi\epsilon_0} [\log_e r]_{r_0}^r$$

$$\log_e \frac{r}{r_0} = \frac{-2\pi\epsilon_0}{\lambda} [V(r) - V(r_0)]$$

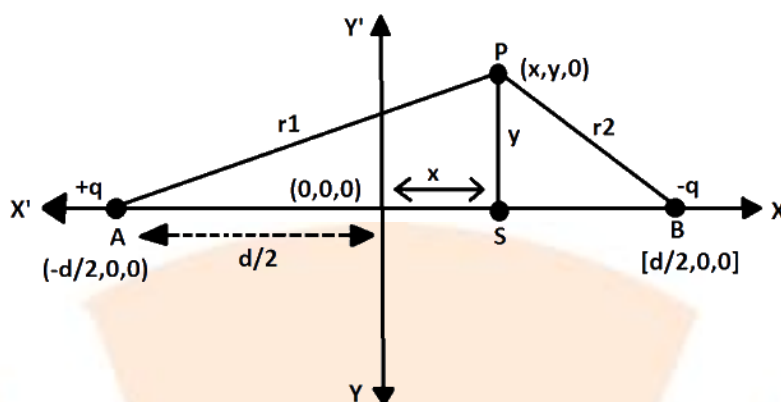
$$\frac{r}{r_0} = e^{\frac{-2\pi\epsilon_0}{\lambda} [V(r) - V(r_0)]}$$

$$r = r_0 e^{\frac{-2\pi\epsilon_0}{\lambda} [V(r) - V(r_0)]}$$

So the equipotential surfaces are the coaxial curved surfaces of cylinders with given cylinder of radius r related as above.

2.25 Two point charges of magnitude $+q$ and $-q$ are placed at $\left(-\frac{d}{2}, 0, 0\right)$ and $\left(\frac{d}{2}, 0, 0\right)$ respectively. Find the equation of the equipotential surface where the potential is zero.

Ans: The potential due to charge $+q$ and $-q$ will be zero in between the line joining the two charges. Let zero potential is at S.



Then the equipotential surface will pass through S and perpendicular to line joining the two charges or AB.

So, $r_1^2 = AS^2 + SP^2$

$$r_1^2 = \left(x + \frac{d}{2}\right)^2 + y^2$$

$$r_1 = \sqrt{\left(x + \frac{d}{2}\right)^2 + y^2}$$

Similarly, $r_2 = \sqrt{\left(x - \frac{d}{2}\right)^2 + y^2}$

So net potential at P = 0

$$\frac{kq}{r_1} + \frac{k(-q)}{r_2} = 0 \quad \text{where } k = \frac{1}{4\pi\epsilon_0}$$

$$\Rightarrow kq \left[\frac{1}{r_1} - \frac{1}{r_2} \right] = 0 \quad [\because kq \neq 0]$$

$$\Rightarrow \frac{1}{r_1} - \frac{1}{r_2} = 0$$

$$\Rightarrow \frac{1}{r_1} = \frac{1}{r_2} \Rightarrow r_1 = r_2$$

$$\Rightarrow \left(x + \frac{d}{2}\right)^2 + y^2 = \left(x - \frac{d}{2}\right)^2 + y^2$$

$$\Rightarrow \left(x + \frac{d}{2}\right)^2 = \left(x - \frac{d}{2}\right)^2$$

$$\Rightarrow x^2 + \frac{d^2}{4} + dx = x^2 + \frac{d^2}{4} - dx$$

$$2dx = 0$$

$$2d \neq 0$$

$$\therefore x = 0$$

So the equipotential surface will be perpendicular to x -axis and passing through $x=0$ i.e., origin in $Y-Z$ plane.

2.26 A parallel plate capacitor is filled by a dielectric whose relative permittivity varies with the applied voltage (U) as $\epsilon = \alpha U$ where $\alpha = 2V^{-1}$. A similar capacitor with no dielectric is charged to $U_0 = 78V$. It is then connected to the uncharged capacitor with the dielectric. Find the final voltage on the capacitors.

Ans: Let C be the capacitance of capacitor C_1 without dielectric then charge $q_1 = CU$ where U is the final potential of the C_1 when connected to C_2 the capacitor filled with dielectric ϵ_0

$$C_2 = \epsilon C$$

$$q_2 = \epsilon CU$$

$$= \alpha UCU$$

Initial charge q_0 of C_1 when charged at potential of $U_0 = 78V$ is,

$$q_0 = CU_0 = 78C$$

By the law of conservation of charge

$$q_0 = q_1 + q_2$$

$$78C = CU + \alpha \cdot CU^2$$

$$78 = U + \alpha U^2 \quad [\alpha = 2 \text{ per volt}]$$

$$\therefore U + 2U^2$$

$$\text{Or } 2U^2 + U - 78 = 0$$

$$U = \frac{-1 \pm \sqrt{1 - 4 \cdot 2(-78)}}{2 \cdot 2} = \frac{-1 \pm \sqrt{1 - 624}}{4}$$

$$U = \frac{-1 \pm \sqrt{625}}{4} = \frac{-1 \pm 25}{4}$$

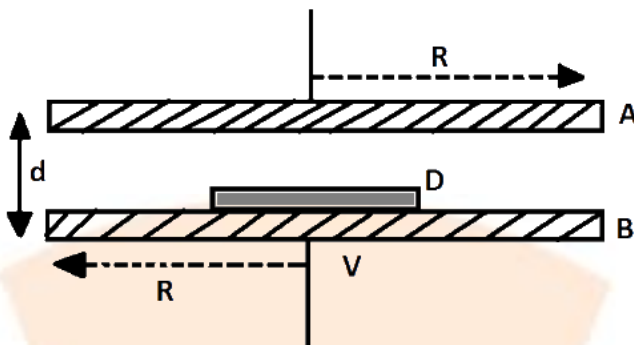
$$U = \frac{-1 \pm 25}{4} \text{ as } U \text{ is positive}$$

$$U = \frac{24}{4} = 6 \text{ volts}$$

The final potential on both the capacitors will be 6volts .

2.27 A capacitor is made of two circular plates of radius R each, separated by a distance $d \ll R$. The capacitor is connected to a constant voltage. A thin conducting disc of radius $r \ll R$ and thickness ' t ' $\ll r$ is placed at a center of the bottom plate. Find the minimum voltage required to lift the disc if the mass of the disc is m .

Ans: Let A and B are circular plates of radius R separated by distance $d \ll R$ kept horizontally.



A thin conducting disc D of radius $r \ll R$ of thickness ' t ' is placed concentrically on lower plate B as shown in figure.

Let plate A and B charged with potential V .

The magnitude of electric field E between plates of capacitor is

$$E = \frac{V}{d} \left[\because E = \frac{-dV}{dr} \right]$$

Consider a Gaussian surface along circular disc D .

$$\text{By Gauss's law, } \oint E \cdot ds = \frac{q'}{\epsilon_0}$$

$$\frac{V}{d} \pi r^2 = \frac{q'}{\epsilon_0}$$

q' is the charge conducted by plate B to disc D during charging. Nature of charge on plate B and disc will be same so repulsive force acts between B and D .

$$\text{So, the charge on disc } D = q' = \frac{V}{d} \pi r^2 \epsilon_0$$

Electrostatic repulsive force acting on disc in upward direction

$$F = q' E = \frac{V}{d} \pi r^2 \epsilon_0 \cdot \frac{V}{d} = \frac{V^2}{d^2} \pi r^2 \epsilon_0$$

This repulsive force will be balanced by weight mg of disc D .

$$mg = \frac{V^2}{d^2} \pi r^2 \epsilon_0$$

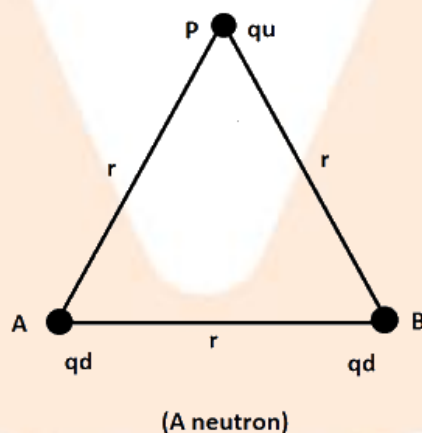
$$V^2 = \frac{mgd^2}{\pi r^2 \epsilon_0}$$

2.28 (a) In a quark model of elementary particles, a neutron is made of one up quarks $\left[\text{charge} \left(\frac{2}{3} \right) e \right]$ and two down quarks $\left[\text{charge} \left(-\frac{1}{3} \right) e \right]$. Assume that they have a triangle configuration with side length of the order of $10^{-15} m$. Calculate electrostatic potential energy of neutron and compare it with its mass $939 MeV$.

(b) Repeat the above exercise for a proton which is made of two up and one down quark.

Ans: (a) $q_d = -\frac{1}{3}e$ [Charge on down quark]

$q_u = +\frac{2}{3}e$ [Charge on up quark]



Potential energy $U = \frac{kq_1q_2}{r}$

Where, $k = \frac{1}{4\pi\epsilon_0}$

$$U = \frac{kq_1q_2}{r} + \frac{kq_1q_3}{r} + \frac{kq_2q_3}{r}$$

$$\therefore U_n = \frac{1}{4\pi\epsilon_0} \frac{(-q_d)(-q_d)}{r} + \frac{(-q_d)q_u}{4\pi\epsilon_0} + \frac{q_u(-q_d)}{4\pi\epsilon_0 r}$$

$$U_n = \frac{q_d}{4\pi\epsilon_0 r} [+q_d - q_u - q_u] \text{ [Talking sign of charge]}$$

$$U_n = \frac{q_d}{4\pi\epsilon_0 r} [+q_d - 2q_u] = \frac{9 \times 10^9 \times \frac{1}{3} e}{10^{-15}} \left[\frac{1}{3} e - 2 \cdot \frac{2}{3} e \right]$$

[Natural sign of charges is already taken]

$$\frac{9 \times 10^9 \times e}{3 \times 10^{-15}} \cdot \frac{e}{3} [1 - 4] \text{ joule}$$

$$\frac{-3 \times 9 \times 10^9 \times 1.6 \times 10^{-19}}{9 \times 10^{-15}} e \text{ joule}$$

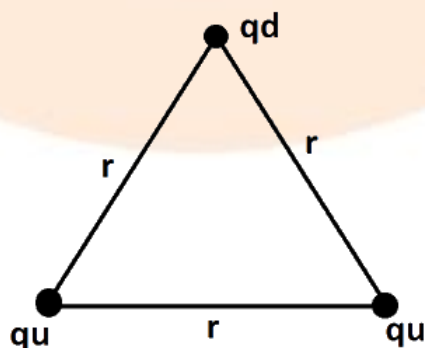
$$U = -0.48 \text{ MeV}$$

So, charges inside neutron [$1q_u$ and $2q_d$] are attracted by energy of 0.48 MeV .

Energy released by a neutron when converted into energy is 939 MeV .

$$\therefore \text{required ratio} = \frac{1 - 0.481 \text{ MeV}}{939 \text{ MeV}} = 0.0005111 = 5.11 \times 10^{-4}$$

(b) P.E. Of proton consists of 2 up and 1 down quark



$$r = 10^{-15} \text{ m}$$

$$q_d = -\frac{1}{3}e, q_u = \frac{2}{3}e$$

$$U_p = \frac{1}{4\pi\epsilon_0} \frac{q_u \times q_u}{r} + \frac{q_u(-q_d)}{4\pi\epsilon_0 r} + \frac{q_u(-q_d)}{4\pi\epsilon_0 r}$$

$$U_p = \frac{q_u}{4\pi\epsilon_0 r} [q_u - q_d - q_d]$$

$$U_p = \frac{q_u}{4\pi\epsilon_0 r} [q_u - 2q_d] = \frac{9 \times 10^9}{10^{-15}} \frac{2}{3} e \left[\frac{2}{3} e - 2 \cdot \frac{1}{3} e \right] = 0$$

2.29 Two metal spheres, one of radius R and the other of radius $2R$, both have same surface charge density σ . They are brought in contact and then separated. What will be new surface charge densities on them?

Ans: Let the surface charge density of both the spheres be σ and their charges are q_1 and q_2

$$\therefore q_1 = \sigma \cdot A_1 = \sigma \cdot 4\pi R^2$$

$$q_2 = \sigma \cdot A_2 = \sigma \cdot 4\pi (2R)^2 = \sigma \cdot 4\pi R^2 \cdot 4 = 4q_1$$

Both charged spheres are kept in contact, so charge flows between them and their potential will become equal.

Let the charges on them now be q_1' and q_2' .

$$\text{So, } V_1 = V_2 \left(\because V = \frac{kq}{r} \right)$$

$$\text{So } \frac{kq_1}{R} = \frac{kq_2}{2R} \left[\because k = \frac{1}{4\pi\epsilon_0} \right]$$

Where q_1' and q_2' are the charges on spheres after redistribution of charges

$$\frac{q_1}{R} = \frac{q_2}{2R}$$

$$\therefore q_2' = 2q_1' \dots\dots\dots \text{I}$$

By the law of conservation of charges

$$q_1 + q_2 = q_1' + q_2'$$

$$q_1 + 4q_1 = q_1' + 2q_1' \text{ (From equation I)}$$

$$5q_1 = 3q_1'$$

$$q_1 = \frac{20}{3} \pi R^2 \sigma$$

$$\sigma_1 = \frac{q_1}{A_1} = \frac{\frac{20}{3} \pi R^2 \sigma}{4\pi R^2} = \frac{5}{3} \sigma$$

$$\sigma_1 = \frac{q_2}{A_2} = \frac{2 \cdot q_1}{4\pi (2R)^2} = \frac{2 \cdot \frac{20}{3} \pi R^2 \sigma}{4\pi \cdot 4R^2} = \frac{5}{6} \sigma$$

$$\text{Hence, } \sigma_1 = \frac{5}{3} \sigma \text{ and } \sigma_2 = \frac{5}{6} \sigma$$

2.30 In the circuit shown in Fig. 2.7, initially K_1 is closed and K_2 is open. What are the charges on each capacitor?

Then K_1 was opened and K_2 was closed (order is important), What will be the charge on each capacitor now? [$C = 1\mu F$]

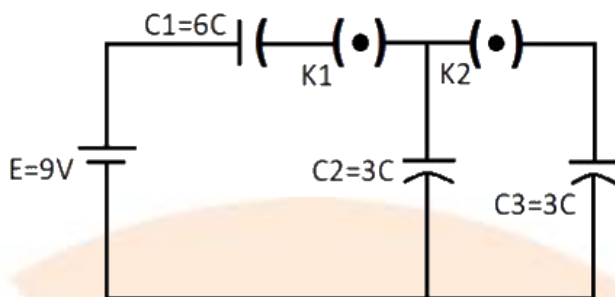


Fig. 2.7

Ans: When K_2 is open and K_1 is closed the capacitors C_1 and C_2 will charge and potential will develop across them i.e. V_1 and V_2 respectively which will be equal to the potential of battery.

$$\therefore V_1 + V_2 = 9 \text{ I}$$

$$\therefore V = \frac{q}{C}$$

So, $V \propto \frac{1}{C}$ and

$$\frac{V_1}{V_2} = \frac{C_2}{C_1}$$

$$\frac{V_1}{V_2} = \frac{3C}{6C}$$

$$3V_2 = 6V_1$$

$$V_2 = 2V_1 \text{ II}$$

From Equations I and II

$$V_1 + 2V_1 = 9$$

$$3V_1 = 9$$

$$V_1 = 3 \text{ volts}$$

$$V_2 = 2 \times 3 \text{ volt}$$

$$= 6 \text{ volt}$$

$$\therefore q_1 = C_1 V_1$$

$$= 6C \times 3$$

$$= 18C \text{ [From II equation, } C = 1\mu F \text{]}$$

$$\text{So, } = 18 \times 1\mu F$$

$$= 18\mu C$$

$$\therefore q_2 = C_2 V_2$$

$$= 3C \times 6$$

$$= 3 \times 1\mu F \times 6$$

$$= 18\mu C$$

So, the charges on each capacitor is $q_1 = q_2 = 18\mu C$

When k_1 is open and k_2 is closed then charge q_2 will be distributed among C_2 and C_3 . Let it be q_2' and q_3

$$\therefore q_2 = q_2' + q_3$$

$$18\mu C = 3 \times 1\mu F \times V + 3 \times 1\mu F \times V$$

$$18 = 6V$$

$$V = 3 \text{ volts}$$

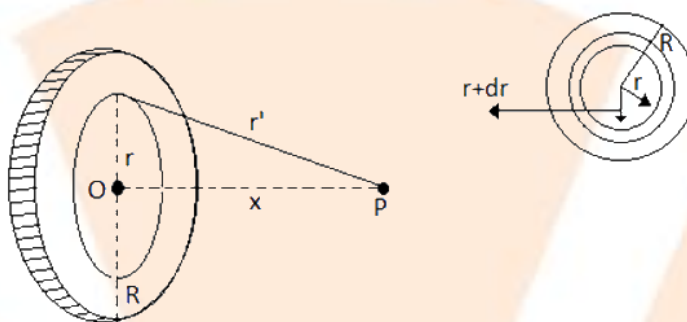
So the potential on C_2 and C_3 capacitors are 3volts each

$$q_2 = C_2 V = 3 \times 1\mu F \times 3 \text{ volt} = 9\mu C$$

$$q_3 = C_3 V = 3 \times 1\mu F \times 3 \text{ volt} = 9\mu C$$

2.31 Calculate potential on the axis of a disc of radius R due to a charge Q , uniformly distributed on its surface.

Ans: Consider a point P on the axis perpendicular to the plane of disc and at a distance x from the center O of disc as shown in the figure.



Now consider a ring of radius r , thickness dr and disc of radius R , as shown in figure. Let the charge on the ring be dq .

The potential dV due to the ring at P , will be

$$dV = \frac{k dq}{r'} \left[\because r' = \sqrt{r^2 + x^2} \right]$$

The charge on the ring is

$$dq = \sigma \cdot \text{area of ring}$$

$$dq = \sigma \cdot \left[\pi (r + dr)^2 - \pi r^2 \right]$$

$$dq = \sigma \cdot \pi \left[r^2 + dr^2 + 2rdr - r^2 \right]$$

As dr is very small, the value of dr^2 will be negligible.

$$\therefore dq = \sigma \pi (2rdr)$$

$$dq = 2\pi r \sigma \cdot dr$$

$$\therefore dV = \frac{k \cdot 2\pi r \sigma dr}{\sqrt{(r^2 + x^2)}}$$

So the potential due to the charged disc is

$$\int_0^V dV = \int_0^R \frac{k \cdot 2\pi r \sigma dr}{\sqrt{(r^2 + x^2)}}$$

$$V = k \cdot 2\pi \sigma \int_0^R \frac{r dr}{(r^2 + x^2)^{\frac{1}{2}}}$$

$$V = 2\pi k \sigma \int_0^R r \cdot (r^2 + x^2)^{\frac{1}{2}} dr$$

$$V = 2\pi k \sigma \left[(R^2 + x^2)^{\frac{1}{2}} - x \right]$$

$$V = \frac{2\pi \sigma}{4\pi \epsilon_0} \left[(R^2 + x^2)^{\frac{1}{2}} - x \right]$$

$$\left[\sigma = \frac{Q}{\pi R^2} \right]$$

$$V = \frac{2\pi R^2 \sigma}{4\pi \epsilon_0 R^2} \left[\sqrt{R^2 + x^2} - x \right]$$

$$V = \frac{2Q}{4\pi \epsilon_0 R^2} \left[\sqrt{R^2 + x^2} - x \right]$$

2.32 Two charges q_1 and q_2 are placed at $(0,0,d)$ and $(0,0,-d)$ respectively. Find the locus of points where the potential is zero.

Ans: Let the potential at any point $P(x, y, z)$ is zero then $-V_1 + V_2 = 0$

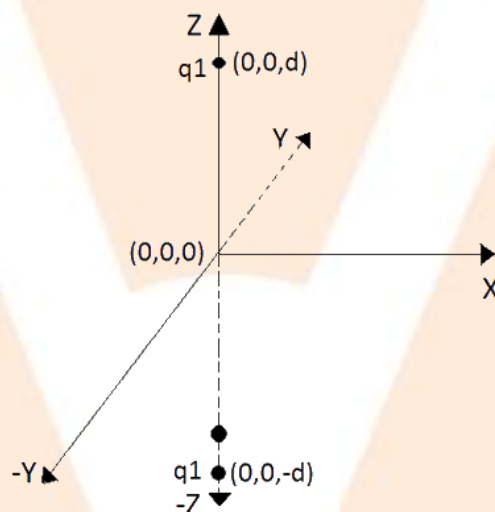
$$\frac{kq_1}{\sqrt{x^2 + y^2 + (z-d)^2}} + \frac{kq_2}{\sqrt{x^2 + y^2 + (z+d)^2}} = 0$$

$$\frac{q_1}{\sqrt{x^2 + y^2 + (z-d)^2}} + \frac{q_2}{\sqrt{x^2 + y^2 + (z+d)^2}} = 0$$

$$\frac{q_1}{\sqrt{x^2 + y^2 + (z-d)^2}} + \frac{-q_2}{\sqrt{x^2 + y^2 + (z+d)^2}}$$

$$\frac{q_1}{q_2} = \frac{-\sqrt{x^2 + y^2 + (z+d)^2}}{\sqrt{x^2 + y^2 + (z-d)^2}}$$

$$\frac{q_1^2}{q_2^2} = \frac{x^2 + y^2 + z^2 + d^2 - 2zd}{x^2 + y^2 + z^2 + d^2 + 2zd}$$



Componendo and dividendo of $\frac{x}{a} = \frac{y}{b}$ is $\frac{x+a}{x-a} = \frac{y+b}{y-b}$

Then componendo and dividendo of

$$\frac{\left(\frac{q_1}{q_2}\right)^2}{1} = \frac{x^2 + y^2 + z^2 + d^2 - 2zd}{x^2 + y^2 + z^2 + d^2 + 2zd}$$

$$\frac{\left(\frac{q_1}{q_2}\right)^2 + 1}{\left(\frac{q_1}{q_2}\right)^2 - 1} = \frac{x^2 + y^2 + z^2 + d^2 - 2zd + (x^2 + y^2 + z^2 + d^2 + 2zd)}{x^2 + y^2 + z^2 + d^2 - (x^2 + y^2 + z^2 + d^2 + 2zd)}$$

$$\left[\frac{\left(\frac{q_1}{q_2}\right)^2 + 1}{\left(\frac{q_1}{q_2}\right)^2 - 1} \right] = \frac{2(x^2 + y^2 + z^2 + d^2)}{-4dz}$$

$$x^2 + y^2 + z^2 + d^2 = -2dz \left[\frac{\left(\frac{q_1}{q_2}\right)^2 + 1}{\left(\frac{q_1}{q_2}\right)^2 - 1} \right]$$

$$x^2 + y^2 + z^2 = 2d \left[\frac{\left(\frac{q_1}{q_2}\right)^2 + 1}{\left(\frac{q_1}{q_2}\right)^2 - 1} \right] z + d^2 = 0$$

$$x^2 + y^2 + z^2 = 2d \left[\frac{q_1^2 + q_2^2}{q_1^2 - q_2^2} \right] z + d^2 = 0$$

This is the equation of sphere with center (a, b, c) as required point is on z -axis so

$$a = 0, b = 0 \text{ and } z = 2d \left[\frac{q_1^2 + q_2^2}{q_1^2 - q_2^2} \right]$$

$$\left(0, 0, -2d \left[\frac{q_1^2 + q_2^2}{q_1^2 - q_2^2} \right] \right)$$

2.33 Two charges $-q$ each are separated by distance $2d$. A third charge $+q$ is kept at mid-point 'O'. Find the potential energy of $+q$ as a function of small

distance x from 'O' due to $-q$ charges. Sketch P.E. v/s x and convince yourself that the charge at O is in an unstable equilibrium.

Ans: $V = \frac{kq}{r}$

Let the equilibrium of $+q$ is P at a distance x from the mid-point of line joining two charges.

Force F_A on $+q$ is towards left side and force F_B is towards right side, so for equilibrium of $+q$ at P ,

$$F_A = F_B$$

$$\frac{-kq^2}{(d-x)^2} = \frac{-kq^2}{(d+x)^2}$$

$$(d-x)^2 = (d+x)^2$$

$$d-x = d+x \text{ (Taking square root)}$$

$$-2x = 0$$

$$x = 0$$

So, Equilibrium position of charge $+q$ between two $-q$ charges is at mid-point (O) of line joining the two charges $-q$ and $-q$.

Now we have to find out potential energy of $+q$ as a function of small distance x from balance condition (O) towards any of $-q$ charge.

Let new position of charge $+q$ from a small distance x from (O)

$$U = \frac{k(q)(-q)}{(d-x)} + \frac{k(q)(-q)}{(d+x)} \left(\because U = \frac{kq_1q_2}{r_1} \right)$$

$$U = -kq^2 \left[\frac{1}{(d-x)} + \frac{1}{(d+x)} \right]$$

$$U = -kq^2 \left[\frac{d+x+d-x}{(d-x)(d+x)} \right] = -kq^2 \left[\frac{2d}{d^2-x^2} \right]$$

$$U = \frac{-q^2}{4\pi\epsilon_0} \cdot \frac{2d}{(d^2-x^2)}$$

So, U is the P.E. as a function of x .

x	U
0	$\frac{-2k}{d} q^2$
$\frac{d}{2}$	$\frac{4}{3} \left(\frac{-2kq^2}{d} \right)$
$-\frac{d}{2}$	$\frac{4}{3} \left(\frac{-2kq^2}{d} \right)$
$+d$	$-\infty$
$-d$	$-\infty$

